Module 9: Assignment - Arnav Kucheriya

Saturday, March 21, 2026

Module 9 Assignment: Round Robin Scheduling

1. Comparison: Round Robin vs FCFS
First-Come, First-Serve (FCFS)
Advantages:
- Simple to implement
- Low overhead
- Processes are executed in order of arrival

Disadvantages:
- Poor response time for short/interactive jobs
- Convoy effect: short processes wait behind long ones
- Not suitable for time-sharing systems

Round Robin (RR)
Advantages:
- Fair CPU allocation
- Good response time
- Suitable for interactive systems
- Prevents long jobs from monopolizing CPU

Disadvantages:
- More context switching overhead
- Performance depends on time quantum
- Large quantum → behaves like FCFS

Conclusion
Round Robin improves fairness and responsiveness compared to FCFS, especially in systems where multiple users or interactive processes are involved.

----

2. Round Robin Scheduling (q = 4)
Given Processes

Process	Arrival Time	Burst Time
P1	0	15
P2	7	13
P3	9	21
P4	41	9
P5	31	8
P6	35	7
P7	51	14

I/O Events
- t = 14 → I/O for 14 units
- t = 16 → I/O for 14 units

- t = 38 → I/O for 7 units
- t = 61 → I/O for 7 units

A) Gantt Chart

0–4 P1
4–8 P1
8–12 P2
12–14 P1
14–16 P3
16–20 P2
20–24 P2
24–25 P2
25–28 IDLE
28–32 P1
32–36 P3
36–38 P5
38–39 P1
39–43 P6
43–47 P3
47–51 P4
51–54 P6
54–58 P5
58–61 P3
61–65 P7
65–69 P4
69–71 P5
71–75 P7
75–79 P3
79–80 P4
80–84 P7
84–88 P3
88–90 P7

B) Turnaround Time (TAT)

Formula:
TAT = Completion Time — Arrival Time

Process	Completion	Arrival	TAT
P1	39	0	39
P2	25	7	18
P3	88	9	79
P4	80	41	39
P5	71	31	40
P6	54	35	19
P7	90	51	39

C) Waiting Time (WT)

Formula:
WT = TAT - Burst - I/O

Process	TAT	Burst	I/O	WT

P1	39	15	14	10
P2	18	13	0	5
P3	79	21	21	37
P4	39	9	0	30
P5	40	8	7	25
P6	19	7	0	12
P7	39	14	0	25

D) Response Time (RT)
Formula:
RT = First CPU Time - Arrival Time

Process	**First Run**	**Arrival**	**RT**
P1	0	0	0
P2	8	7	1
P3	14	9	5
P4	47	41	6
P5	36	31	5
P6	39	35	4
P7	61	51	10

Waiting Time
P1 = 10, P2 = 5, P3 = 37, P4 = 30, P5 = 25, P6 = 12, P7 = 25
Response Time
P1 = 0, P2 = 1, P3 = 5, P4 = 6, P5 = 5, P6 = 4, P7 = 10
Turnaround Time
P1 = 39, P2 = 18, P3 = 79, P4 = 39, P5 = 40, P6 = 19, P7 = 39